ANDRÉ NICOLAS SILVA

Software Engineer | Product, AI, Systems & Software Engineering

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PROFESSIONAL SUMMARY

Software Engineer with a **product and problem-solving focus**, with hands-on experience building and evolving production SaaS products used by **10+ companies**. Works across product requirements, technical decisions, software architecture, frontend, backend, APIs, data, AI, cloud infrastructure, debugging, and production delivery.

Combines software engineering with an **AI-native development workflow**, using AI as an engineering multiplier to accelerate implementation, explore solutions, understand codebases, review changes, debug issues, generate tests, and iterate on product functionality while maintaining technical judgment, architecture, correctness, and maintainability.

Strong experience with **TypeScript, Angular, React, Golang, REST APIs, SQL, PostgreSQL, concurrency, event-driven systems, Server-Sent Events (SSE), CI/CD, and Cloudflare**, with the ability to move across application layers according to the problem being solved.

Takes ownership of problems rather than individual tasks, translating **business requirements and product needs into working software** while balancing UX, business rules, technical constraints, performance, reliability, and delivery.

CORE COMPETENCIES

Product & Problem Solving	Product Thinking, problem decomposition, requirements analysis, business rules, UX, feature ownership, scope definition, technical decision-making, independent delivery
AI-Native Engineering	Generative AI, LLM integration, prompt engineering, contextual prompting, structured AI responses, AI-assisted problem solving, AI-assisted coding, AI-driven development workflows
Software Engineering	Full-stack development, system design, software architecture, debugging, code review, automated testing, maintainability, reliability, production support
Architecture & Systems	Software Architecture, Hexagonal Architecture, Component-Based Architecture, frontend architecture, separation of concerns, reusable systems, maintainable system boundaries
Backend & APIs	Golang, Node.js, Java, Spring/Spring Boot, REST APIs, internal APIs, backend services, concurrency, asynchronous processing, event-driven systems, Server-Sent Events (SSE), service integrations, automations
Frontend & Web	TypeScript, JavaScript, Java (MSPA and academic Android/SQLite), Node.js (MSPA and project work), Angular, React, Next.js, Angular Signals, RxJS, Web Components, Lit, Shadow DOM, Tailwind CSS
Data	SQL, PostgreSQL, Supabase, relational databases, database migrations, persistence, queries
Security & Integrity	Input validation and sanitization, authorization and access control, privacy-aware consent flows, audit events, secure product safeguards
AI Product Engineering	Genkit, LLM-powered features, input sanitization, contextual information, structured outputs, reliable AI-powered product functionality
AI Development Stack	Codex, Claude, Cursor, CodeRabbit, Skills, MCPs, Agents, SDD, AI harnesses, deterministic evaluation, scope, output-quality, and model-cost guardrails
Cloud & Infrastructure	Cloudflare Workers, KV, D1, Durable Objects, AWS/Azure cloud services, Podman/Docker-compatible containers, Bash, Testcontainers, production Docker images, CI/CD, Git
Performance & Quality	Performance optimization, SEO, Lighthouse, Vitest, automated testing, code review, debugging, production reliability

PROFESSIONAL EXPERIENCE

MSPA — Frontend Engineer (full-stack contributions) *Apr 2025 – Present*

SaaS platform for Privacy, Compliance (LGPD), and Digital Auditing

- **Product Engineering & Problem Ownership:** Built and evolved core SaaS products, including Compass and Trust Center, used by **10+ companies**, taking ownership from product context and requirements through technical definition, implementation, refinement, debugging, and production delivery.
- **Product & Technical Decision-Making:** Translated business requirements and product needs into technical solutions, contributing to feature scope, business rules, UX, architecture, implementation strategy, and technical trade-offs.
- **AI-Native Engineering:** Applied an AI-driven development workflow using **Codex, Claude, Cursor, CodeRabbit, Skills, MCPs, Agents, and SDD** to accelerate implementation, explore solutions, understand codebases, debug issues, review changes, and iterate on product functionality while maintaining architectural consistency and code quality.
- **Generative AI Product Engineering:** Integrated **Generative AI into production SaaS products**, developing user-facing capabilities with contextual prompt engineering, input sanitization, and structured AI responses using **Genkit**, with emphasis on reliable and consistent product functionality.
- **Full-Stack Engineering:** Implemented production functionality across frontend, backend, APIs, data, and infrastructure, moving between application layers according to product requirements and technical problems.
- **Expanded Platform Stack:** Contributed at MSPA with **Node.js, Java, Spring/Spring Boot, AWS/Azure cloud services, Podman/Docker-compatible containers, Bash automation, Testcontainers, production Docker images, testing, accessibility, technical documentation, UI delivery, and security-oriented engineering**, selecting the appropriate layer according to product requirements.
- **Backend & Systems Engineering:** Contributed to **Golang backend services and APIs**, investigating and implementing solutions involving concurrency, asynchronous processing, events, service integrations, automations, and production issues across the application stack.
- **Data & Business Logic:** Worked with **SQL and PostgreSQL**, including queries, data persistence, database migrations, business rules, and investigation of data-related issues across application workflows.
- **Software Architecture:** Participated in architectural decisions and implemented solutions following **Hexagonal Architecture**, component-based architecture, separation of responsibilities, reusable systems, and maintainable system boundaries.
- **Web & Edge Systems:** Developed an agnostic consent engine using **Web Components, Lit, and Shadow DOM**, integrated with **Cloudflare Workers, KV, D1, and Durable Objects** for edge-based execution and distributed functionality.
- **Performance & Reliability:** Implemented performance and SEO improvements, investigating bottlenecks and optimizing loading behavior and application performance while maintaining usability and production reliability.
- **Engineering Delivery:** Worked across the software delivery lifecycle, including **CI/CD, deployment, debugging, production investigation, and integration of changes across application layers**.
- **Engineering Coordination:** Reorganized development workflow during a critical project phase, improving scope clarity, sprint organization, and productivity across a **6-person team**, with direct coordination of 4 team members.

OPEN SOURCE CONTRIBUTIONS

Stoat / Revolt Platforms LTD — Volunteer Frontend Engineer *2026 – Present*

Contribute to the open-source project with a focus on **UX engineering and frontend development**, identifying user-facing problems and turning them into technical improvements that make the product clearer and more intuitive for users. **Pull request approved by project maintainers.**

PROJECTS

RendeCerto — Investment-scenario simulator *Next.js, TypeScript, Vitest*

Built a financial simulation and calculation application focused on investment scenarios, translating financial scenarios into deterministic business rules and calculation logic with automated testing to validate results across different scenarios.

Terto Beats — Digital-assets marketplace for music *Next.js, TypeScript, Supabase, Vercel*
Built an end-to-end digital commerce platform for selling digital products, translating product and business requirements into the web interface, checkout, payment state, data persistence, integrations, and post-purchase automations.

Quebrando Fronteiras — Social-impact and donation platform *Next.js, TypeScript*
Refactored critical user flows by identifying UX and product-flow problems and turning them into technical improvements focused on clarity and conversion.

Marca Racha — Local match discovery and coordination *Next.js, React, Supabase, Prisma, Resend, Zod, Vitest*
Implemented group goals, invitations, and declared contributions with authentication and real-time data. Behavioral tests verify query-boundary ownership, role-derived capabilities, membership-aware visibility, notification deduplication, and status transitions.

Saúde em Campo — Offline-first mobile architecture *React Native, Expo, SQLite, Drizzle, TanStack Query*
Designed an application for teams recording field visits and health information even without connectivity. The proposal combines per-user local isolation, a sync queue, idempotency, retries/backoff, RBAC, and risk prioritization; it is architecture evidence, not a clinical-deployment claim.

Missões do Dia — Android application *Java, SQLite, SharedPreferences, Services, BroadcastReceiver, AsyncTask, XML*
Built an academic Android mission-CRUD application with local persistence, native components, and XML layouts.

EDUCATION

Tecnólogo in Internet Systems — UNCISAL (Brazil) *Completed 2026*

LANGUAGES

Portuguese: Native ▪ **English:** Advanced